

Zirui Wang

Ph.D. Candidate, Materials Science and Engineering
University of Illinois Urbana-Champaign

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SUMMARY: Ph.D. candidate in Materials Science with 5+ years of experience in electrochemical diagnostics and Python-based laboratory automation. Skilled in developing automated experimental platforms, optimizing electrochemical processes, and translating materials characterization insights into scalable engineering and manufacturing decisions.

TECHNICAL SKILLS

Materials and Electrochemical Characterization: XRD, Raman, IR, SEM, EDS, ellipsometry, goniometry, four-point probe, profilometry, AFM, CV, EIS, galvanostatic and potentiostatic methods, differential capacity analysis (dQ/dV), Tafel analysis, bulk electrolysis, potentiometry, chronoamperometry, scanning electrochemical microscopy

Software and Data Processing: Arbin, CHI, COMSOL, Digi-Elch, Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), MATLAB, Java, L^AT_EX, HTML, GitHub, Arduino, Raspberry Pi, ImageJ, Vesta, Klayout, Fusion 360, Adobe Illustrator, Blender, Internet of Things (IoT), Data curation, regression/classification, Bayesian optimization, PCA, computer vision

Fabrication and Materials Processing: Glovebox, vacuum oven, CAD, microfabrication, photolithography, thin-film deposition (electrodeposition, electroplating, spin coating, e-beam, sputtering, CVD), etching (reactive-ion, wet etching), microfluidics, bonding, device packaging, 3D printing

EDUCATION

University of Illinois Urbana-Champaign 2022 - 2027 (Expected)
Ph.D. Materials Science (Advisor: Prof. Joaquín Rodríguez-López) *GPA: 3.96/4.00*

Thesis research: *High-Throughput Characterization and Optimization of Electrochemical Energy Storage Systems Guided by Automated Electrochemistry Platforms*

University of Illinois Urbana-Champaign 2018 - 2022
B.Sc., Materials Science with high honor (Research Advisor: Prof. Yingjie Zhang) *GPA: 3.93/4.00*

RESEARCH AND ENGINEERING EXPERIENCE

Automated, High-Throughput Electrochemistry Platform for Evaluating Battery Materials 2022 - Now

- Led end-to-end development of an automated electrochemical characterization platform to rapidly evaluate battery materials.
- Designed and microfabricated interdigitated electrode arrays and developed a Python-controlled experimental automation system integrating potentiostats, solution-handling robots, custom PCBs, and Arduino microcontrollers, reducing manual operations by 80% and increasing experimental throughput by 5×.
- Conducted process optimization and standardization, improving experimental reproducibility and robustness by >20%.

Accelerated Characterization and Optimization of Next-Generation Battery Materials 2023 - Now

- Applied automated platform for high-throughput electrochemical testing of electrodeposited battery materials. (polypyrrole, azo polymers, Prussian Blue) across diverse electrolyte systems, completing >500 experiments in <50 hours per material.
- Quantified interfacial kinetics, chemical reversibility, and energy density by coupling experiments with finite-element modeling (COMSOL) to guide experimental interpretation.
- Validated the structural-property relationship of battery materials via SEM, EDS, XRD, Raman, and profilometry.

Platform Scale-up and Deployment for Device-Level Validation 2025 - Now

- Translated high-throughput screening results into device-level validation by designing and performing flow battery cycling tests using electrolytes optimized by the automated platform.
- Transferred and deployed the automated electrochemical platform to external collaborators (Argonne, UChicago, PNNL), enabling high-throughput characterization across multiple electrochemical systems.

Precise Surface Profiling at Nanometer Scale Enabled by Computer Vision and Machine Learning 2021-2023

- Developed a convolutional neural network to correct tip-convolution artifacts in AFM, improving nanoscale imaging accuracy.
- Generated synthetic datasets and validated reconstruction on experimental nanoparticle images.

SELECTED PUBLICATIONS AND CONFERENCES

- [1] Precise Surface Profiling at the Nanoscale Enabled by Deep Learning. **Nano Letters**, 2024 [\[Link\]](#) [\[Journal Cover\]](#)
- [2] Automated Screening of Electrocatalyst for Electro-oxidation and Feedstock Valorization. **ACS Catalysis**, 2024 [\[Link\]](#)
- [3] Automated Evaluation of Polymeric Battery Materials **ACS Measurement Science Au**, 2026 [\[Link\]](#) [\[Journal Cover\]](#)
- [4] Automated Characterization of Prussian Blue Redox-Flow Batteries **Oral Presentation, 248th ECS Meeting** [\[Link\]](#)
- [5] Accelerated Optimization of Energy Materials for Flow Batteries **Oral Presentation, Batteries GRS, 2026** [\[Link\]](#)

LEADERSHIP AND OUTREACH

Student mentoring: 3 graduate students and 3 undergraduate students. **Workshop:** Instructor for the **Electrochemistry Bootcamp** (Urbana, IL, 2024-2026, 120 participants) **Training:** Attendee, **Next Generation Electrochemistry (NGenE) Workshop** (Argonne National Laboratory, 2025, 40 graduate students) **Outreach:** Presenter, Beckman Institute Open House (Urbana, IL, 2023-2026, >600 visitors/year) **Teaching:** Teaching assistant, MSE 307 (Materials Laboratory I, 30 students)

AWARDS

Beckman Institute Graduate Fellow (2025-2026), **Excellent teaching assistant** (MSE 307, Fall 2024), **Best poster award** (2024 American Vacuum Society Symposium), **Departmental scholarships** (Robert Bohl & Henry E. Grein Jr. Scholarships.)